Three-State DMP with Time-Limited UI: Derivations + some results

Setup and notation

States:

$$E \text{ (employed)}, \qquad I \text{ (unemployed insured)}, \qquad U \text{ (unemployed uninsured)}.$$

Transitions:

$$E \xrightarrow{\delta} I, \qquad I \xrightarrow{\lambda} U, \qquad I, U \xrightarrow{f(\theta)} E,$$

where

$$f(\theta) \equiv \theta q(\theta).$$

Flow payoffs:

$$w \text{ in } E, \qquad b_I \text{ in } I, \qquad b_U \text{ in } U.$$

In this setting, the outside option in wage bargaining is the value of insured unemployment since workers who separate from employment enter the insured state

Laws of motion

The laws of motion are:

$$\frac{de}{dt} = f(\theta)\left(u_I + u_U\right) - \delta e, \tag{1}$$

$$\frac{du_I}{dt} = \delta e - (\lambda + f(\theta))u_I, \tag{2}$$

$$\frac{du_U}{dt} = \lambda u_I - f(\theta)u_U. \tag{3}$$

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Steady-state

In steady state,

$$\frac{de}{dt} = \frac{du_I}{dt} = \frac{du_U}{dt} = 0.$$

From (1):

$$\delta e = f(\theta)(u_I + u_U). \tag{4}$$

From (2):

$$\delta e = (\lambda + f(\theta))u_I. \tag{5}$$

From (3):

$$\lambda u_I = f(\theta)u_U. \tag{6}$$

1 Bellman equations

$$rV_E = w + \delta\,(V_I - V_E), \tag{7}$$
$$rV_I = b_I + f(\theta)\,(V_E - V_I) + \lambda\,(V_U - V_I), \tag{8}$$
$$rV_U = b_U + f(\theta)\,(V_E - V_U). \tag{9}$$

2 Surplus computations

2.1 Difference between U and I: compute $V_U - V_I$

Subtract (9) from (8):

$$rV_I - rV_U = (b_I - b_U) + f(\theta)\big[(V_E - V_I) - (V_E - V_U)\big] + \lambda(V_U - V_I).$$

$$\begin{aligned} r(V_I - V_U) &= (b_I - b_U) + f(\theta)\,(V_U - V_I) + \lambda\,(V_U - V_I) \\ &= (b_I - b_U) - \big(f(\theta) + \lambda\big)\,(V_I - V_U). \end{aligned}$$

$$\big(r + f(\theta) + \lambda\big)\,(V_I - V_U) = b_I - b_U.$$

Therefore,

$$\boxed{V_U - V_I = \frac{b_U - b_I}{r + f(\theta) + \lambda}.} \tag{10}$$

2.2 Employment–insured unemployment surplus: compute $V_E - V_I$

Subtract (8) from (7):

$$rV_E - rV_I = (w - b_I) + \delta(V_I - V_E) - f(\theta)(V_E - V_I) - \lambda(V_U - V_I).$$

$$r(V_E - V_I) = (w - b_I) - \delta(V_E - V_I) - f(\theta)(V_E - V_I) - \lambda(V_U - V_I).$$

$$\big(r + \delta + f(\theta)\big)\,(V_E - V_I) = (w - b_I) - \lambda(V_U - V_I).$$

Thus,

$$\boxed{V_E - V_I = \frac{(w - b_I) - \lambda(V_U - V_I)}{r + \delta + f(\theta)}.} \tag{11}$$

Substitute (10) into (11):

$$\boxed{V_E - V_I = \frac{(w - b_I) - \lambda\left(\dfrac{b_U - b_I}{r + f(\theta) + \lambda}\right)}{r + \delta + f(\theta)}.} \tag{12}$$

3 Compute the reservation wage rV_I

Start from (8):

$$rV_I = b_I + f(\theta)(V_E - V_I) + \lambda(V_U - V_I).$$

Substitute (12) and (10):

$$rV_I = b_I + f(\theta)\left[\frac{(w - b_I) - \lambda\left(\dfrac{b_U - b_I}{r + f(\theta) + \lambda}\right)}{r + \delta + f(\theta)}\right] + \lambda\left(\frac{b_U - b_I}{r + f(\theta) + \lambda}\right). \tag{13}$$

For lighter equations define

$$B(\theta) \equiv \frac{b_U - b_I}{r + f(\theta) + \lambda}. \tag{14}$$

Then (13) is

$$rV_I = b_I + f(\theta)\frac{(w - b_I) - \lambda B(\theta)}{r + \delta + f(\theta)} + \lambda B(\theta). \tag{15}$$

4 Compute the Wage-setting equation

Wage setting follows the same form as in the benchmark model :

$$\boxed{w = rV_I + \beta\big(y - rV_I\big) = (1 - \beta)\, rV_I + \beta y.} \tag{16}$$

Substitute rV_I into wage setting

Plug (15) into (16):

$$w = (1 - \beta)\left[b_I + f(\theta)\frac{(w - b_I) - \lambda B(\theta)}{r + \delta + f(\theta)} + \lambda B(\theta)\right] + \beta y. \tag{17}$$

$$w = (1 - \beta)b_I + (1 - \beta)\, f(\theta)\frac{(w - b_I) - \lambda B(\theta)}{r + \delta + f(\theta)} + (1 - \beta)\lambda B(\theta) + \beta y. \tag{18}$$

Séparer the middle fraction:

$$(1 - \beta)\, f(\theta)\frac{(w - b_I) - \lambda B(\theta)}{r + \delta + f(\theta)} = (1 - \beta)\frac{f(\theta)}{r + \delta + f(\theta)}(w - b_I) - (1 - \beta)\frac{f(\theta)}{r + \delta + f(\theta)}\lambda B(\theta). \tag{19}$$

Insert (19) into (18):

$$w = (1 - \beta)b_I + (1 - \beta)\frac{f(\theta)}{r + \delta + f(\theta)}(w - b_I) - (1 - \beta)\frac{f(\theta)}{r + \delta + f(\theta)}\lambda B(\theta) + (1 - \beta)\lambda B(\theta) + \beta y. \tag{20}$$

Factorize by $(1 - \beta)\lambda B(\theta)$ from the two last terms in (20):

$$\begin{aligned} -(1 - \beta)\frac{f(\theta)}{r + \delta + f(\theta)}\lambda B(\theta) + (1 - \beta)\lambda B(\theta) &= (1 - \beta)\lambda B(\theta)\left(1 - \frac{f(\theta)}{r + \delta + f(\theta)}\right) \\ &= (1 - \beta)\lambda B(\theta)\frac{r + \delta}{r + \delta + f(\theta)}. \end{aligned}$$

Thus (20) becomes

$$w = (1-\beta)b_I + (1-\beta)\frac{f(\theta)}{r+\delta+f(\theta)}(w-b_I) + (1-\beta)\lambda B(\theta)\frac{r+\delta}{r+\delta+f(\theta)} + \beta y. \tag{21}$$

Develop the middle term:

$$(1-\beta)\frac{f(\theta)}{r+\delta+f(\theta)}(w-b_I) = (1-\beta)\frac{f(\theta)}{r+\delta+f(\theta)}w - (1-\beta)\frac{f(\theta)}{r+\delta+f(\theta)}b_I.$$

Plugging into (21) gives

$$w = (1-\beta)b_I - (1-\beta)\frac{f(\theta)}{r+\delta+f(\theta)}b_I + (1-\beta)\frac{f(\theta)}{r+\delta+f(\theta)}w + (1-\beta)\lambda B(\theta)\frac{r+\delta}{r+\delta+f(\theta)} + \beta y. \tag{22}$$

Combine the first two b_I terms:

$$\begin{aligned}(1-\beta)b_I - (1-\beta)\frac{f(\theta)}{r+\delta+f(\theta)}b_I &= (1-\beta)b_I\left(1-\frac{f(\theta)}{r+\delta+f(\theta)}\right)\\ &= (1-\beta)b_I\frac{r+\delta}{r+\delta+f(\theta)}.\end{aligned}$$

Hence

$$w = (1-\beta)\frac{r+\delta}{r+\delta+f(\theta)}b_I + (1-\beta)\frac{f(\theta)}{r+\delta+f(\theta)}w + (1-\beta)\frac{r+\delta}{r+\delta+f(\theta)}\lambda B(\theta) + \beta y. \tag{23}$$

Bring the w term to the left:

$$w\left[1-(1-\beta)\frac{f(\theta)}{r+\delta+f(\theta)}\right] = \beta y + (1-\beta)\frac{r+\delta}{r+\delta+f(\theta)}\left[b_I + \lambda B(\theta)\right]. \tag{24}$$

Result : wage-setting curve

Divide both sides of (24) by the bracket:

$$\boxed{w(\theta) = \frac{\beta y + (1-\beta)\dfrac{r+\delta}{r+\delta+f(\theta)}\left[b_I + \lambda\dfrac{b_U - b_I}{r+f(\theta)+\lambda}\right]}{1-(1-\beta)\dfrac{f(\theta)}{r+\delta+f(\theta)}}.} \tag{25}$$

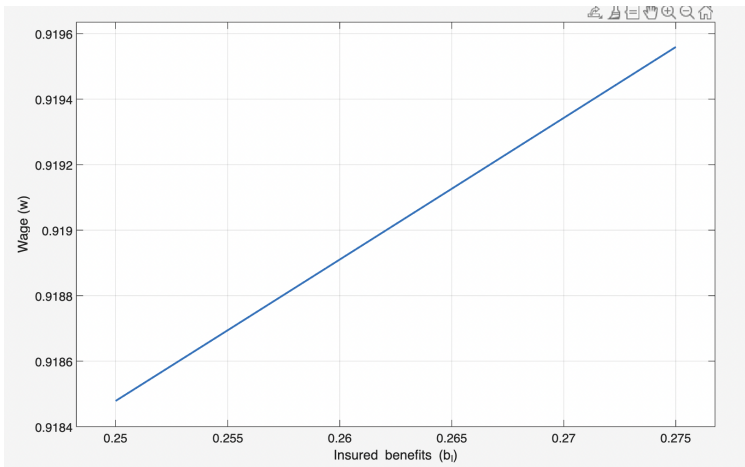
Close the model

To close the model, combine (25) with job creation (same as benchmark model):

$$\boxed{\frac{\gamma}{q(\theta)} = \frac{y-w(\theta)}{r+\delta}}$$

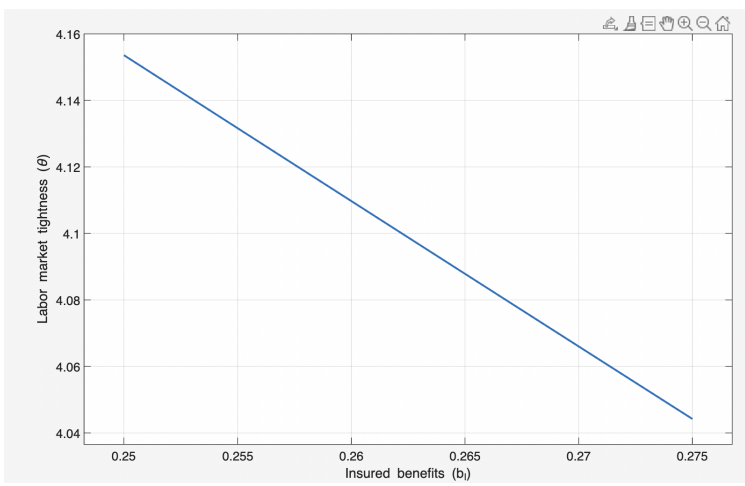
5 How the model solves in matlab

5.1 Higher insured benefits raise wages



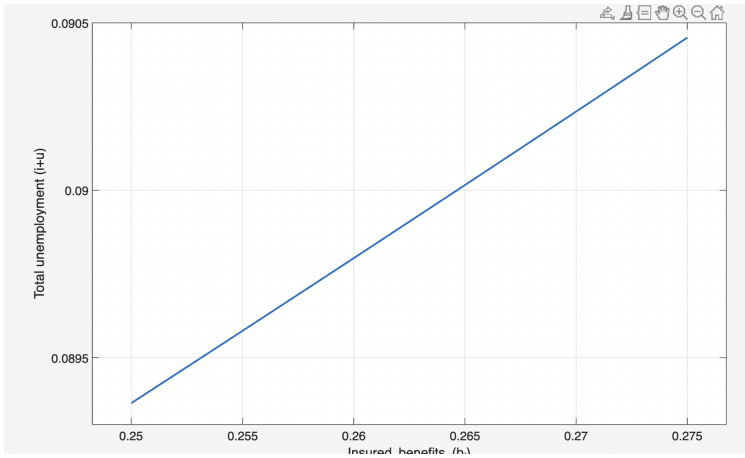
An increase in insured unemployment benefits leads to a monotonic increase in equilibrium wages. Workers capture a larger share of match surplus, reflecting a stronger outside option in wage bargaining.

5.2 Higher insured benefits reduce labor market tightness



As insured benefits increase, labor market tightness falls. Higher wages reduce firms' incentives to post vacancies, leading to fewer vacancies relative to job seekers.

5.3 Higher insured benefits increase total unemployment



Total unemployment rises with insured benefits. The decline in labor market tightness lowers the job-finding rate, increasing unemployment duration and raising the steady-state unemployment rate